Savitribai Phule Pune University
(Formerly University of Pune)

S.Y.B.Sc.(Computer Science)

With

Major: Computer Science

(Faculty of Science and Technology)

(For Colleges Affiliated to Savitribai Phule Pune University)

Choice Based Credit System (CBCS) Syllabus Under National Education Policy (NEP)

To be implemented from Academic Year 2025-2026

Title of the Course: B.Sc.(Computer Science)

Field Project using Software Engineering Techniques

S.Y.B.Sc.(Computer Science)

Major: Computer Science

CS-231-FP Semester III

Field Project

Name: _____

College Name: _____

Roll No.: _____ **Division:** _____

Academic Year: _____

Submission Deadline: / /

S.Y.B.Sc. (Computer Science) Mini Project

Academic Year (20 - 20)

Project Title:_____

Team members:

1) **Name :** _____

Roll No . **Exam Seat No:**

2) **:** _____

Roll No . **Exam Seat No:**

Name _____

Roll No . **Exam Seat No:**

Project Guide Name: _____

Project Guide Signature: _____

Start Date: / / Completion Date: / /

DEPARTMENT OF COMPUTER SCIENCE

CERTIFICATE

This is to certify that **Ms./Mr.** _____
have successfully and satisfactorily completed and submitted the field project titled_____ in partial fulfilment of **S.Y.B.Sc.(CS)** Lab course **CS-231-FP Semester III** prescribed by Savitribai Phule Pune University during the academic year _____.

(Project Guide) (H.O.D. Computer Science)

INTERNAL EXAMINER EXTERNAL EXAMINER

Introduction

The **Online Voting System** is a web-based application designed to make the voting process simple, secure, efficient, and transparent. The system allows registered voters to cast their votes online without the need to visit a physical polling station.

The main purpose of this project is to reduce manual work, minimize paperwork, save time, and provide faster election results. The system can manage voter registration, candidate information, voter authentication, vote casting, and result generation through a centralized platform.

Security is an important part of the system. Only authorized and registered voters should be allowed to vote, and the system should ensure that each voter can cast their vote only once. The results can be generated automatically after the voting process is completed.

Overall, the **Online Voting System** provides a convenient and organized approach to conducting elections while improving efficiency, accuracy, and accessibility.

Preliminary Investigation and its Activity

Problem Identification

- Manual voting takes more time.
- Vote counting can lead to errors.
- Managing election records is difficult.
- Traditional voting requires more manpower.
- Results are not available immediately.

Problem Definition

An online voting system is a web application that allows users to cast their votes electronically. It ensures that each registered user can vote only once and helps in counting votes automatically. Need a secure and efficient online voting system.

Problem Description

1. Traditional voting methods are slow and inefficient.
2. Manual vote counting is time consuming.
3. There is a possibility of human errors during elections.
4. Maintaining voter records manually is difficult.
5. An online system can provide faster & more accurate results
6. The project aims to make voting simple secure and convenient

Fact Finding Technique

1) Interview

We conducted an interview with the concerned person to understand the existing voting process. Information was collected about voter verification, vote casting, vote counting, problems faced during voting, and the requirements of the proposed Online Voting System.

2) Questionnaires

A questionnaire was prepared to collect information from users about the existing voting system. Questions were related to voting difficulties, security, voter authentication, ease of use, and the features expected in an Online Voting System.

3) Document Analysis

Existing documents, voting forms, voter details, and related records were studied to understand how information is collected, stored, and managed. This helped us identify the data required for the proposed system.

4) Observation

The existing offline voting process was observed during the visit. We observed voter verification, the voting procedure, waiting time, staff involvement, and other difficulties. This helped us understand the practical requirements of the new Online

Drawback of Existing System

- Existing voting systems are mostly manual and time-consuming.
- They require significant manpower and resources.
- Vote counting can be prone to human errors.
- *Results are often delayed.*
- Security and transparency may be limited.
- Maintaining voter records manually is difficult.
- There is a possibility of duplicating or invalid voting.
- Existing systems usually lack features such as OTP verification, QR-code login, and Multi language support.

Scope of Proposed System

- Provide secure online voting through a web application.
- Allow users to vote from anywhere with internet access.
- Prevent duplicate voting and ensure one vote per user.
- Generate results automatically and quickly.
- Support OTP verification for secure user authentication.
- Provide QR-code login for easy and fast access.
- Send email and SMS notifications after successful voting.
- Offer multi-language support.
- Can be used for school, college, and organizational elections.
- Can be enhanced with additional security features in the futu

Data Requirement of the System

→ Voter data:

- Voter ID
- Voter Name
- Username & Password
- Mobile Number
- Email Address
- OTP Details
- Language Preference

→ Candidate data:

- Candidate Name
- Candidate ID
- Age
- Gender
- Education
- Status (Won/Lost)
- Symbol
- Party Name
- Vote Result

→ Admin data:

- Admin ID
- Admin Name
- Username

- Password
- Email
- Role

→ Election Data

- Election Type (College / Local / National etc.)
- Date
- Time
- Total Candidates/Voters
- Status (Upcoming / Schedule)

→Election Result Data

- Vote Result
- Candidate Status (Won/Lost)
- No. of Votes to Candidate

Identifying End Users of the System

1. **Voters**– Register, log in, and cast their votes.
2. **Administrator** – Manages voters, candidates, elections and the views of results.
3. **Candidate** – View their profile and election results.
4. **Election`` officers** – Monitor the election process and ensure smooth operation.

Functional Requirements

Functional requirements describe **what the system does.**

1. The admin can log in to the Election Management System using a valid username and password.
2. The admin can add, edit, delete, and view candidate details.
3. S
4. The admin can add, edit, delete, and view candidate details.
5. The admin can view all voter and candidate records.
6. The admin can store and manage election results after the election is completed.
7. The admin can search for voter, candidate, and election records.
8. The admin can generate election reports such as voter lists, candidate lists, and election result reports.

Non-Functional Requirements

Non-functional requirements describe **how the system should work**.

❑ Usability

The system should be easy to use and understand for all users.

❑ Security

The system should allow access only to authorized users through a secure login system.

- **Performance**

The system should respond quickly to user requests and perform operations efficiently.

- **Reliability**

The system should work accurately and consistently without errors.

- **Availability**

The system should be available whenever it is required by the users.

- **Scalability**

The system should handle an increasing number of voters, candidates, and election records.

- **Maintainability**

The system should be easy to maintain and update whenever required.

- **Data Integrity**

The system should ensure that all stored data is accurate, consistent, and secure.

E-R Model

The ER model represents the entities, attributes, and relationships involved in an Online Voting System.

Entity and Attributes

Entity and Attributes

1	**Voter**	Voter_ID,Name,DOB,Voter_Gender,Voter_age,Vote_Address,Voter_Email
2	**Candidate**	C_name , C_ID , C_Age , C_Party , C_Symbol , C_Education , C_Gender
3	**Election**	e_ID , e_starttime , e_endtime , e_Date , e_Address , e_name
4	**Vote**	V_ ID , e_ID , C_ID , Voter_ID
5	**Admin**	A_ID , A_name , A_email , A_password , A_Role , A_phone_no
6	**Result**	R_ID , R_status , C_ID , e_ID , Total_votes

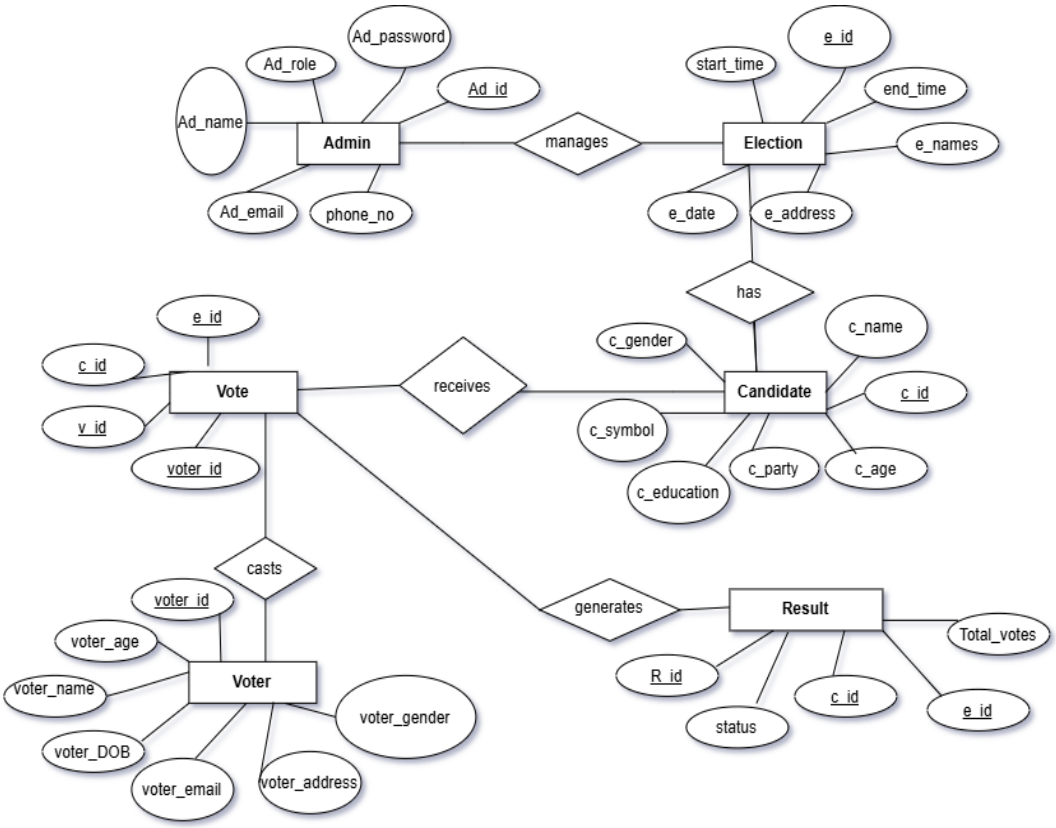
Relationship

Table: Relationships in Online Voting System

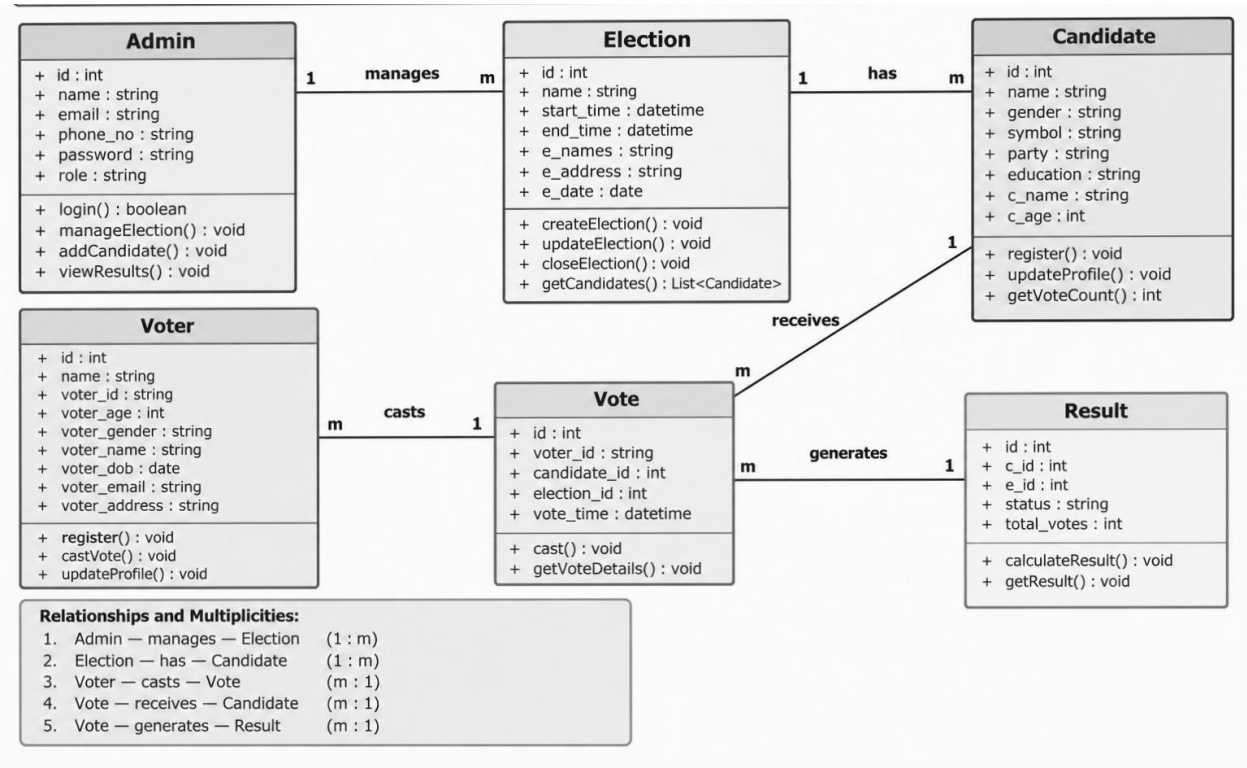
Sr. No.	Entity 1	Relationship	Entity 2
1	Admin	Manages	Election
2	Election	Has	Candidate
3	Voter	Casts	Vote
4	Candidate	Receives	Vote
5	Vote	Generates	Result

Relationship names correspond to the five relationship diamonds in the ER diagram.

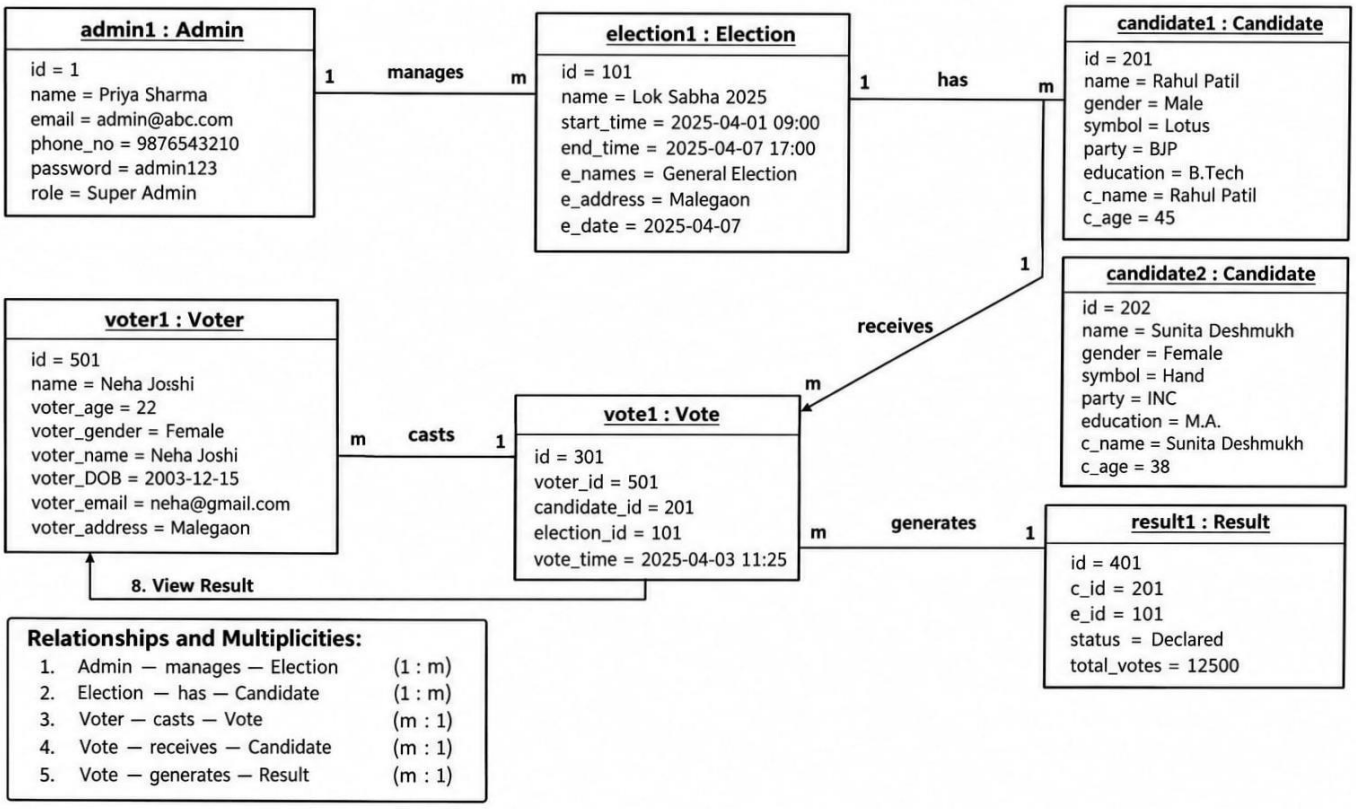
ER-Diagram


Ad_password
Ad_role
Ad_id
Ad_name
Admin
manages
Election
e_id
start_time
end_time
e_names
Ad_email
phone_no
e_date
e_address
has
e_id
c_id
Vote
v_id
voter_id
receives
c_gender
c_name
Candidate
c_id
c_symbol
c_party
c_age
c_education
casts
voter_id
voter_age
voter_name
Voter
voter_gender
voter_DOB
voter_email
voter_address
generates
Result
Total_votes
R_id
c_id
e_id
status

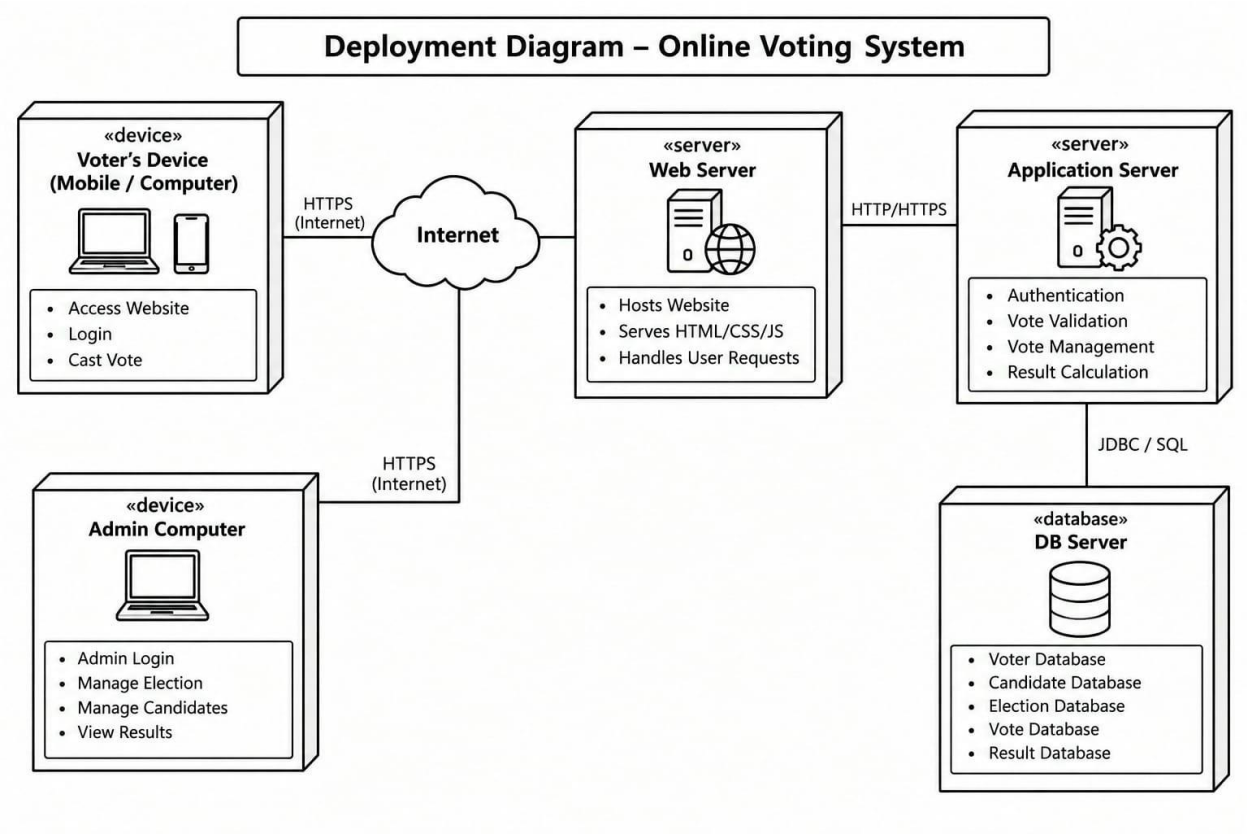
Class Diagram



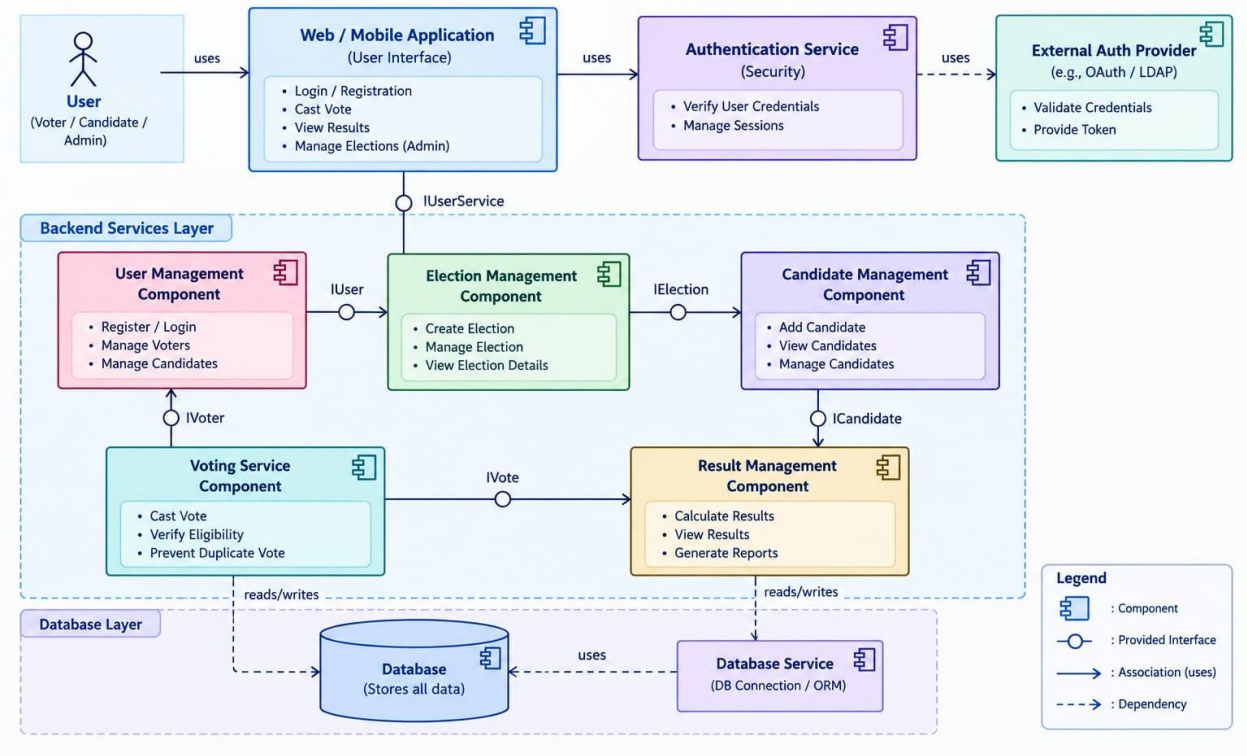
Object Diagram



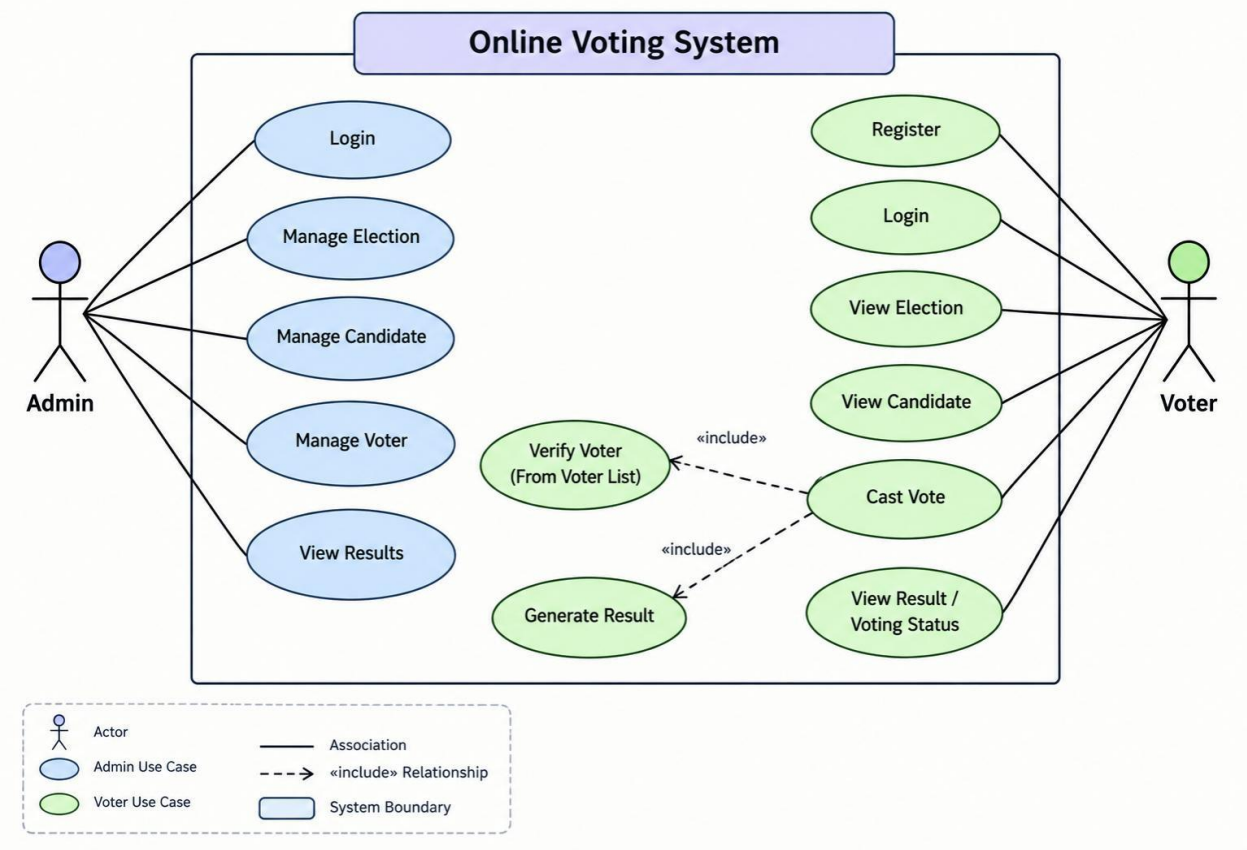
Deployment diagram



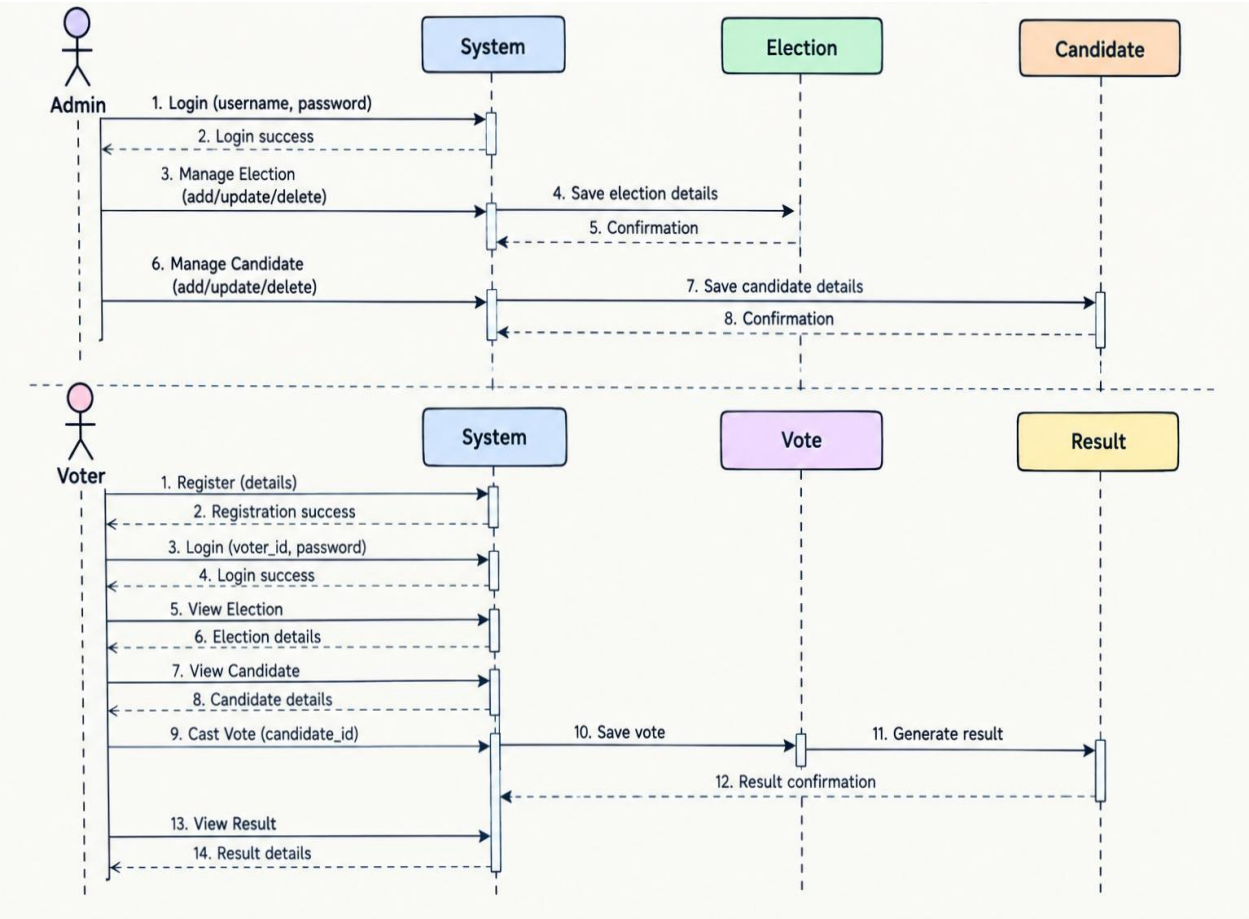
Component Diagram



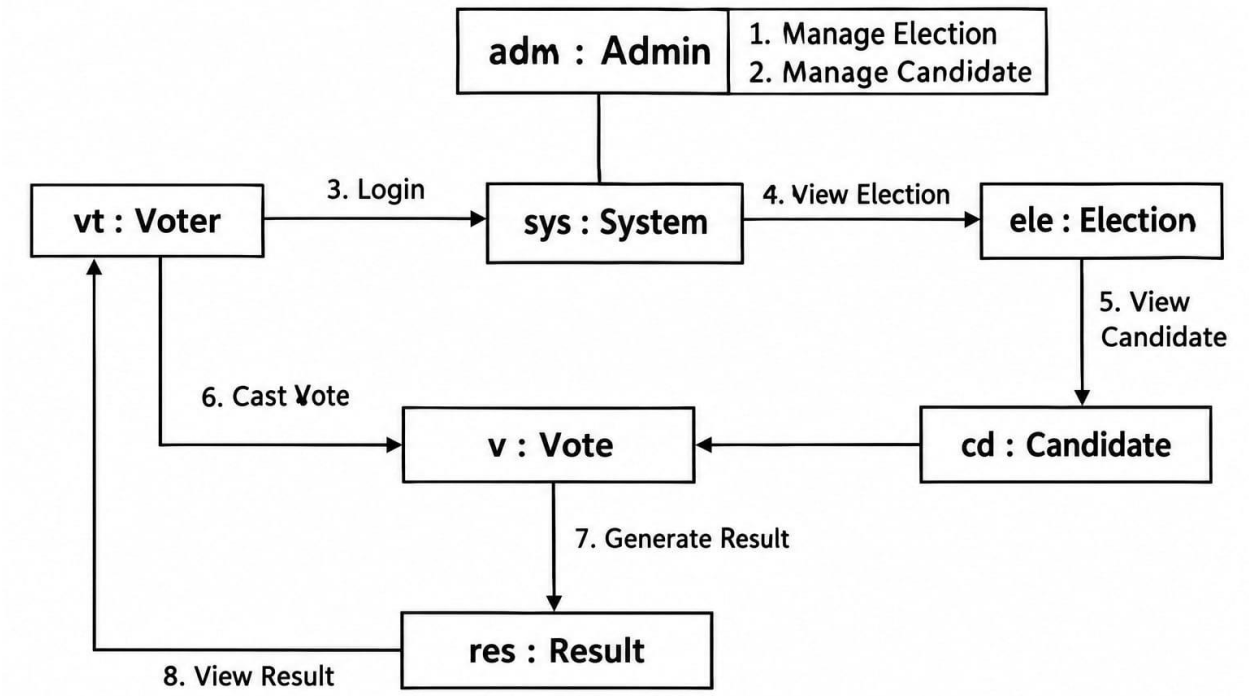
Use Case Diagram



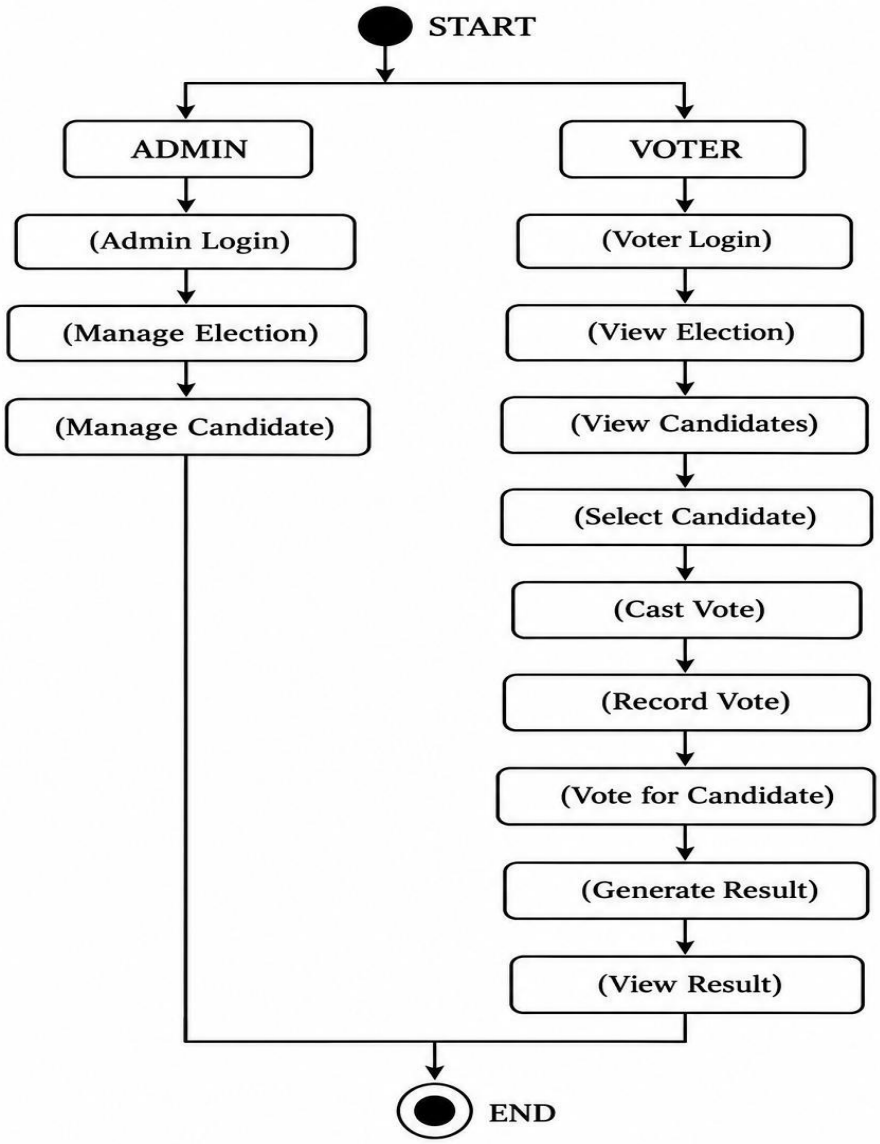
Sequence Diagram



Collaboration Diagram



Activity Diagram



State Chart Diagram

